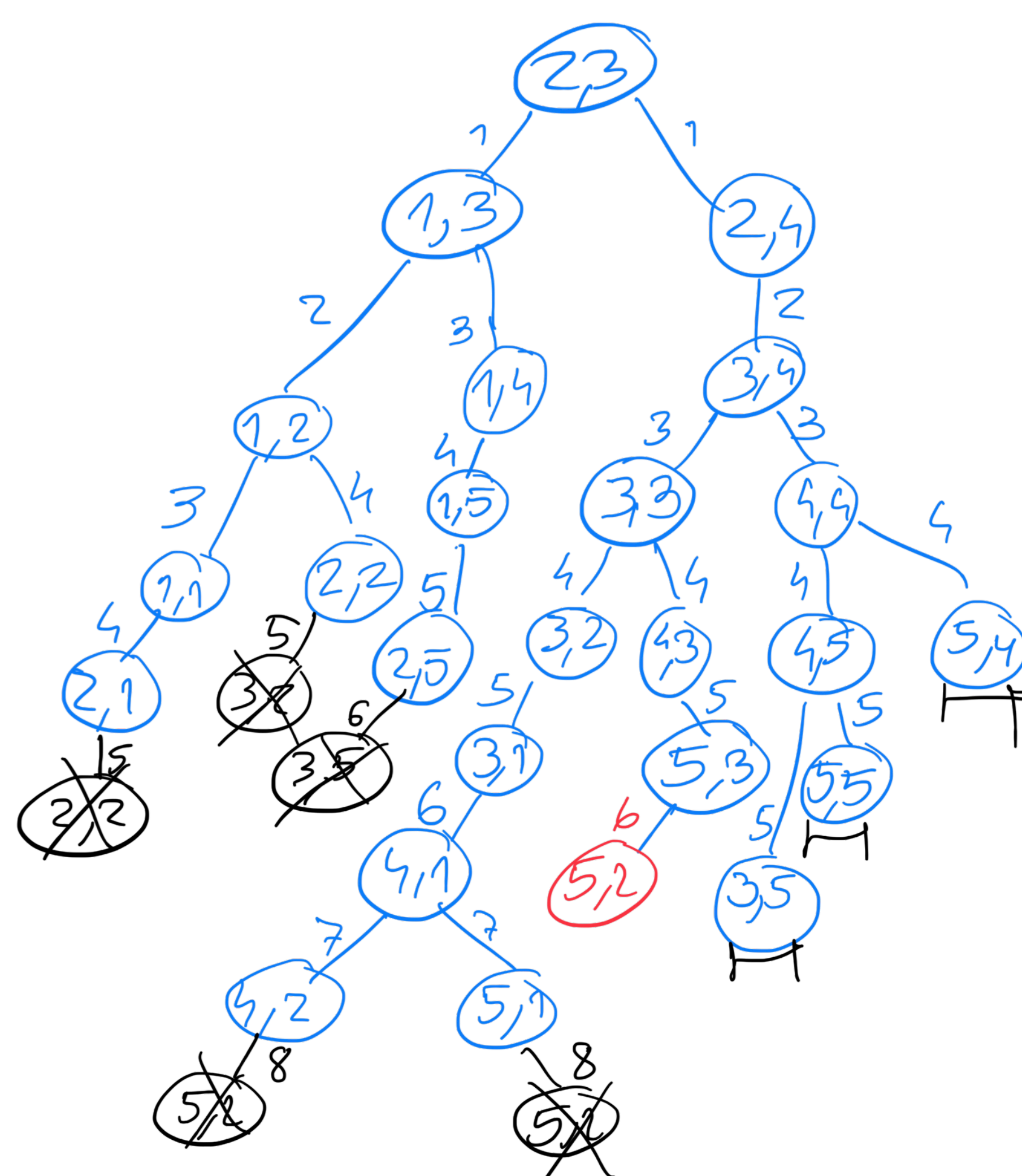
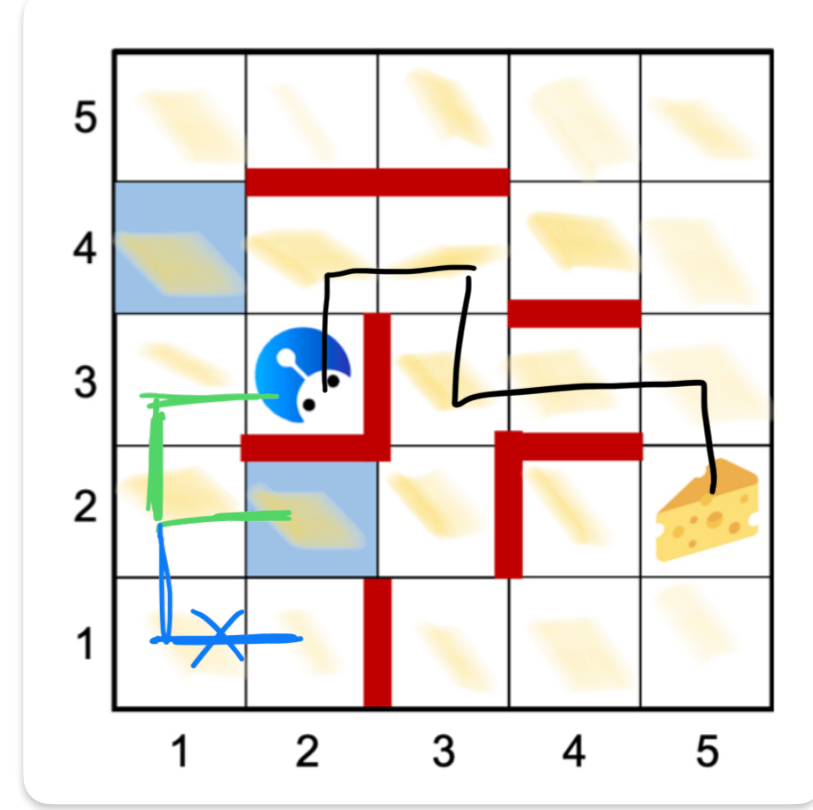
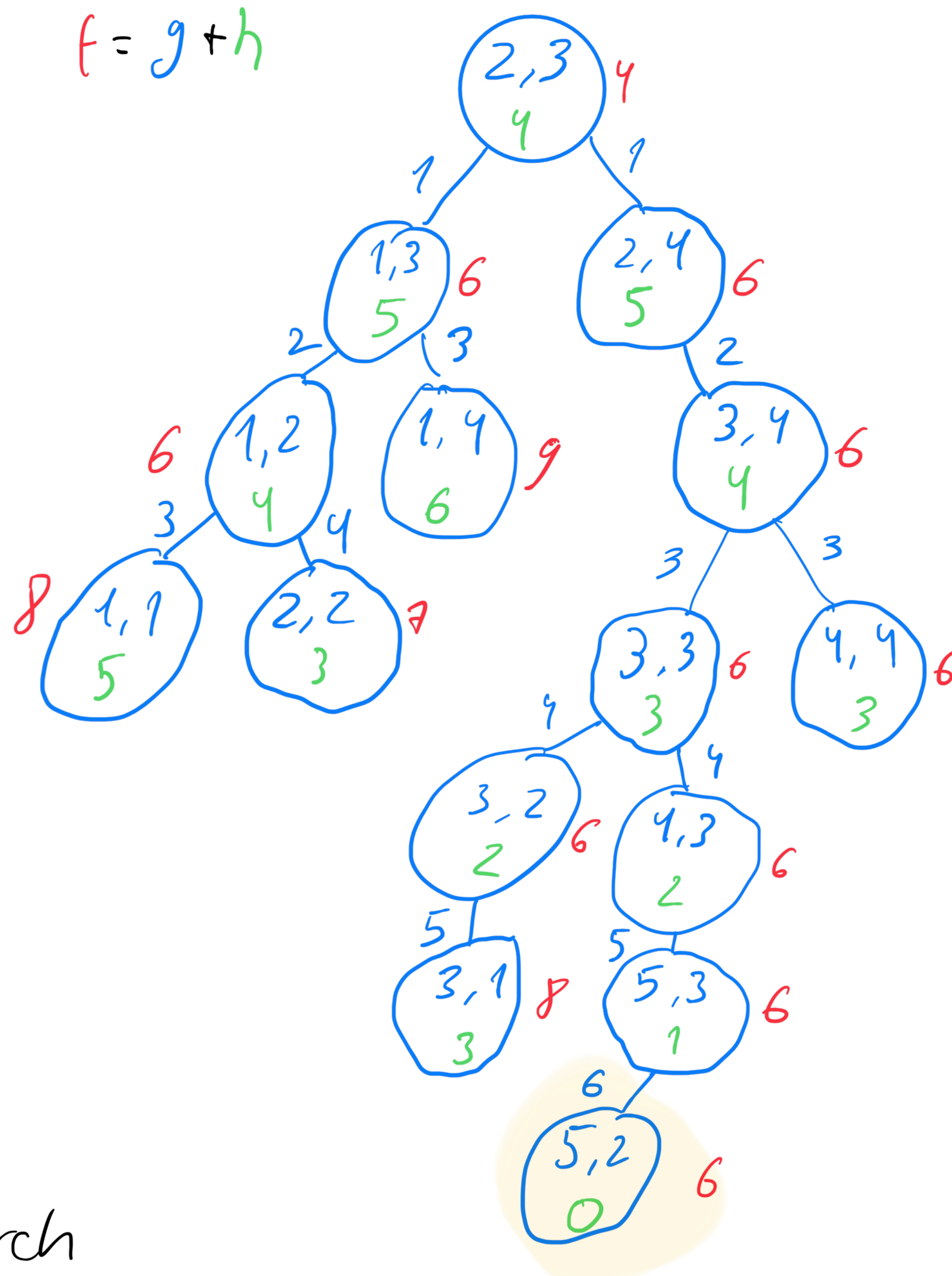
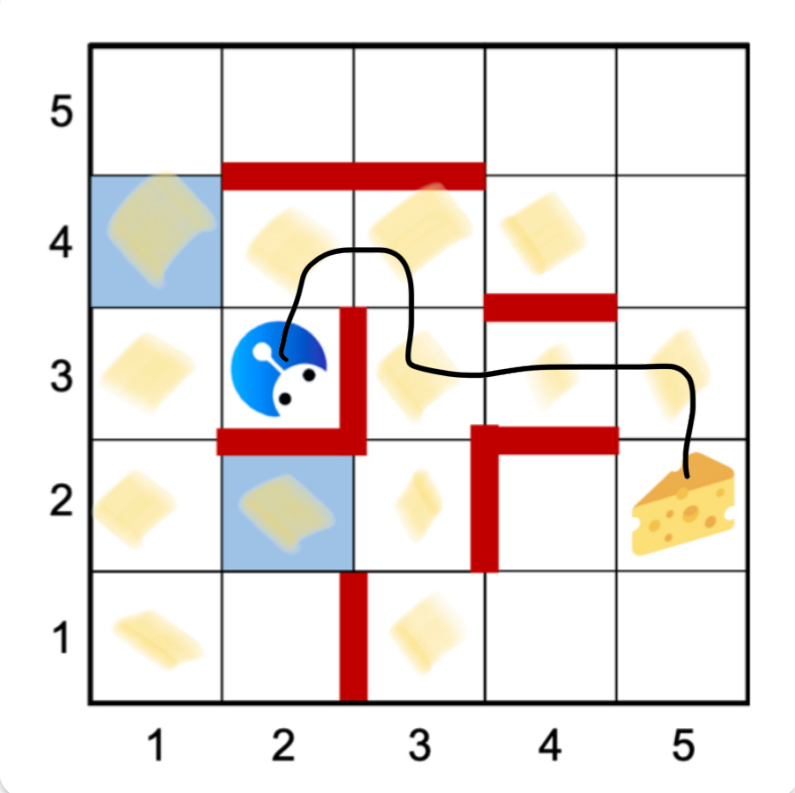


Ü 3.1 a) UCS

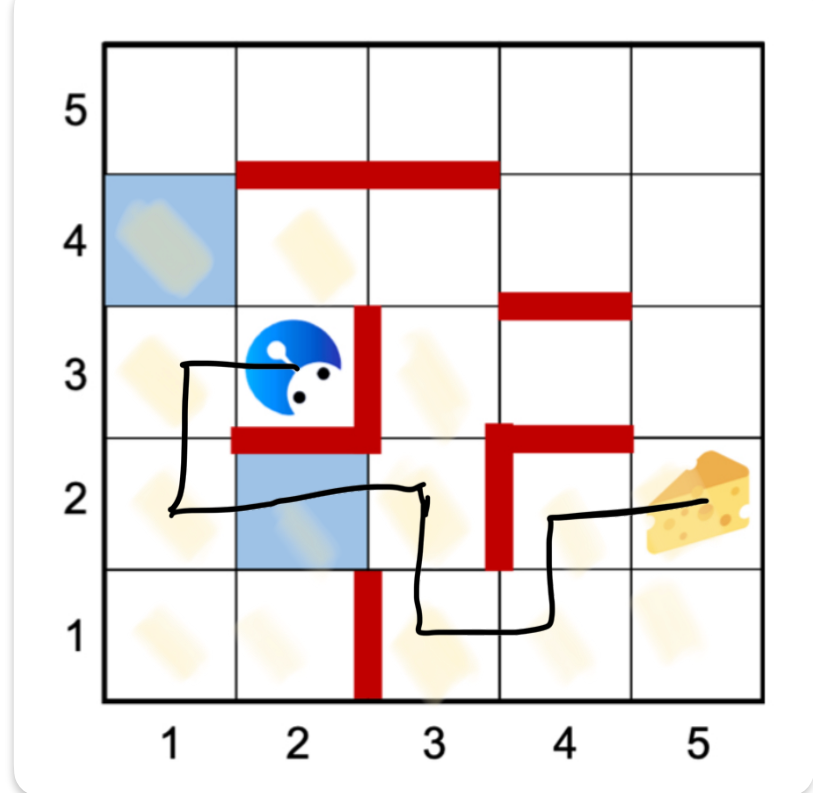


b) A* Search

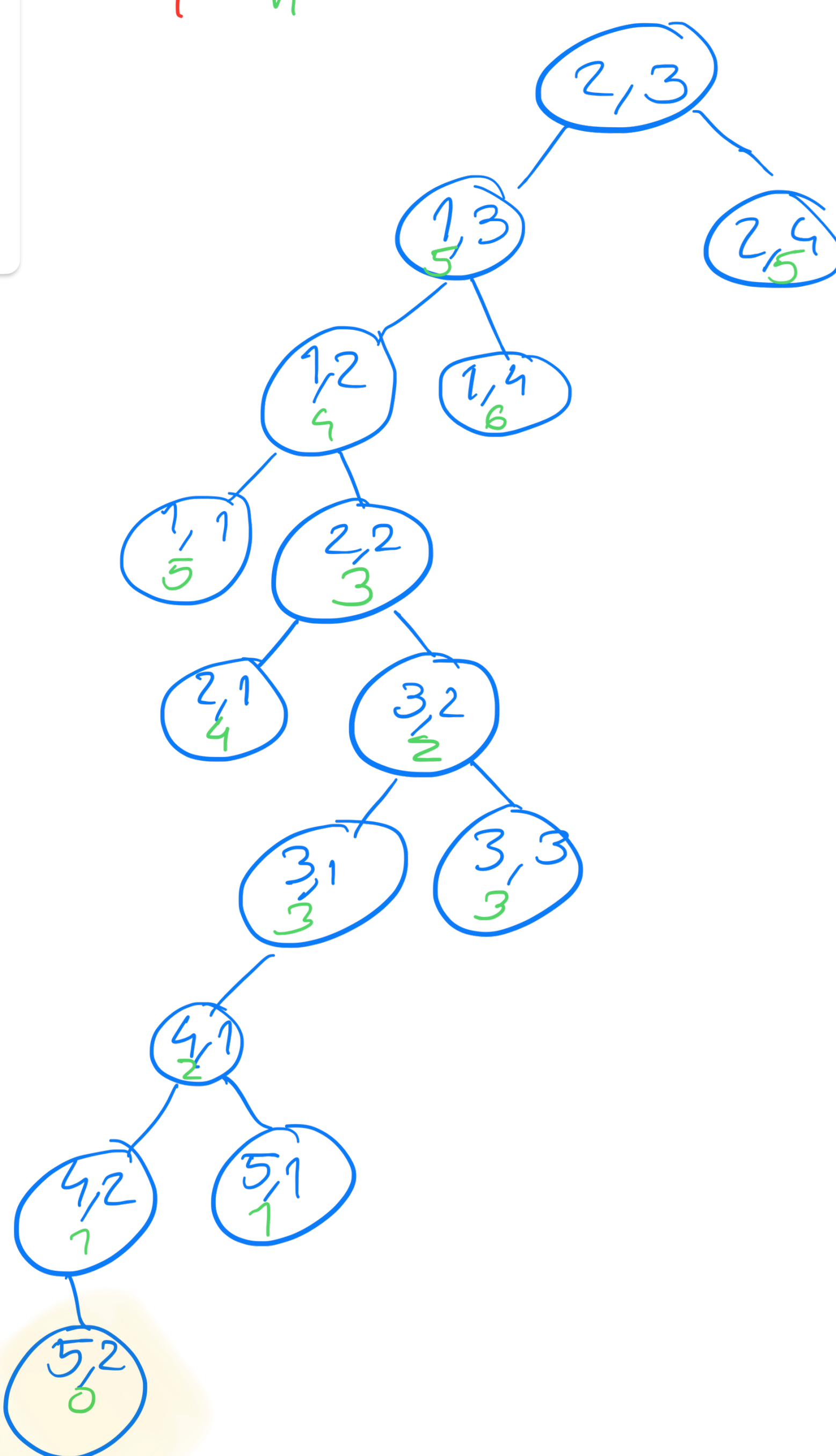
$$f = g + h$$



c) Greedy Search



$$f = h$$



d) UCS: 25 nodes / A* Search: 15 / Greedy Search: 15

UCS and A* Search gave the optimal path

UCS → Was the slowest but gave the optimal path.

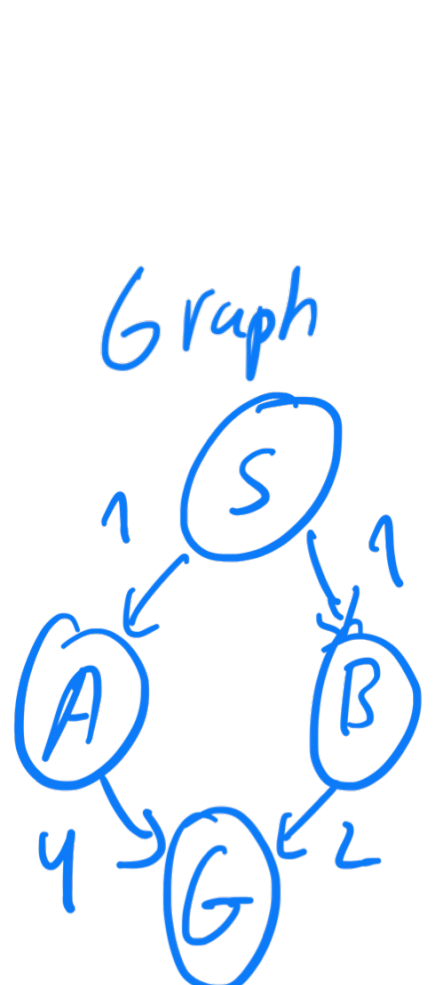
A* Search → Was Fast and gave the optimal path.

Greedy Search → Was Fast but didn't give the optimal path.

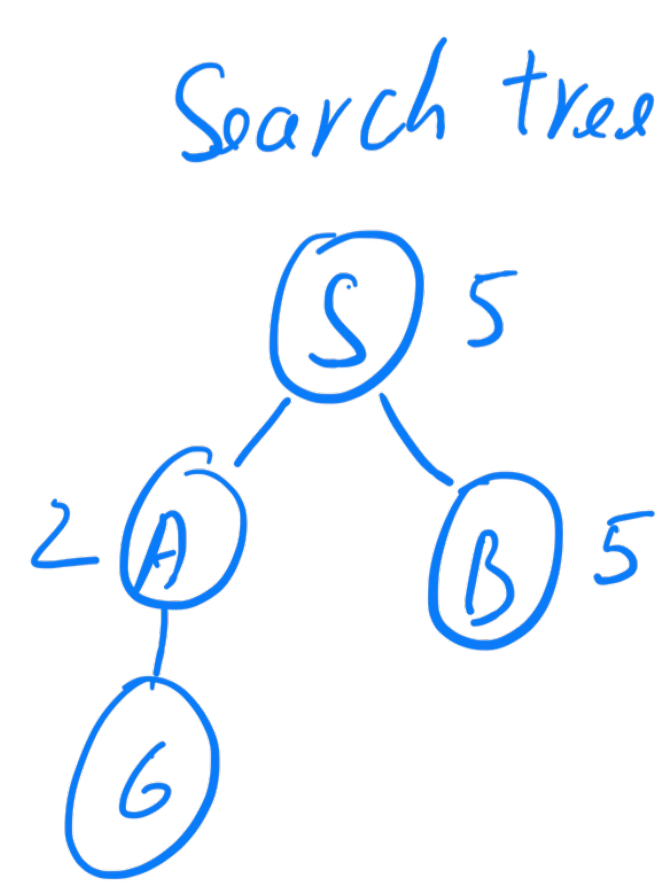
Ü 3.2

It's admissible to use the Euclidean distance as it always underestimates the true remaining travel cost (straight line distance). For this problem, diagonal movements are not admissible, therefore the Manhattan distance is a better heuristic.

Ü 3.3



Heuristics
 $h(S) = 5$
 $h(A) = 1$
 $h(B) = 4$
 $h(G) = 0$



With this example with a bad calculated heuristic the A* Search algorithm returns a not optimal solution.

$S \rightarrow A \rightarrow G$ (Cost = 5)

$S \rightarrow B \rightarrow G$ (Cost = 3)